

LM358 is used as dual comparator to generate a pulse when a visitor enters or leaves.

IR LED and photodiode. The IR LED and IR photodiode pair is used to sense the entry or exit of a visitor.

Two such pairs are used in this project—one at the entry gate and one at the exit gate.

The circuit diagrams of the transmitter and receiver are shown in Fig. 5 and Fig. 6, respectively.

When a visitor enters the hall, he/she momentarily blocks the IR wave transmitted by the IR LED at the entry gate and some energy of the IR wave reflected back from the visitor's body is sensed by the IR photodiode. The LM358 acts as a dual comparator and converts the momentary fluctuation detected by the photodiode to a pulse. The pulse is transmitted to the receiver side by the 434MHz transmitter. The two outputs of the comparator (one for entry and one for exit) are connected to two data pins of HT12E encoder.

After successful wireless transmission, the pulses arrive on the data pins



Fig. 3: Block diagram of the transmitter



Fig. 4: Block diagram of the receiver

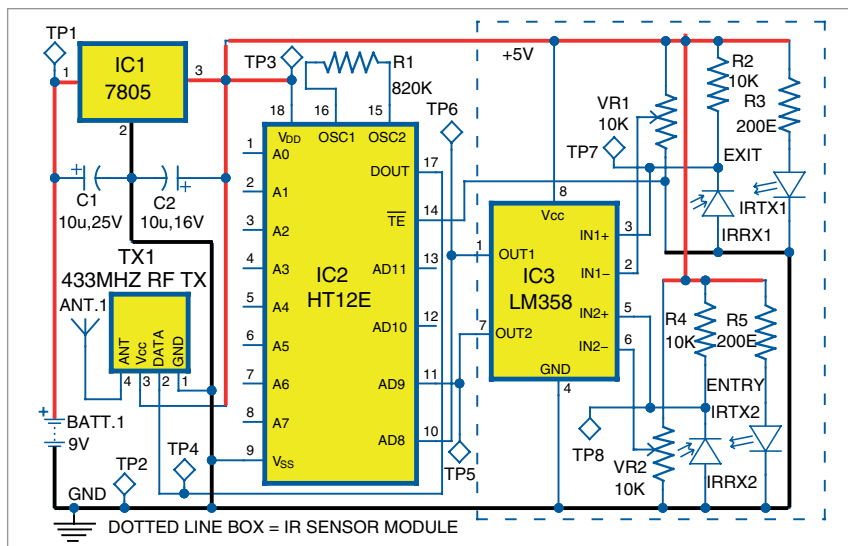


Fig. 5: Circuit diagram of transmitter

PARTS LIST

Semiconductors:

- IC1, IC4 - 7805, 5V voltage regulator
- IC2 - HT12E encoder
- IC3 - LM358 op-amp
- IC5 - HT12D decoder
- IRTX1, IRTX2 - IR transmitting LED
- IRRX1, IRRX2 - IR receiving LED
- Board1 - Arduino Uno

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 820-kilo-ohm
- R2, R4 - 10-kilo-ohm
- R3, R5 - 200-ohm
- R6 - 47-kilo-ohm
- R7 - 220-ohm
- VR1-VR3 - 10-kilo-ohm preset

Capacitors:

- C1, C3 - 10 μ F, 25V electrolytic
- C2, C3 - 10 μ F, 16V electrolytic

Miscellaneous:

- BATT.1, BATT.2 - 9V battery
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module
- LCD1 - 16 $\times$ 2 LCD display
- Jumper wires
- 30 pin berg strips (male)

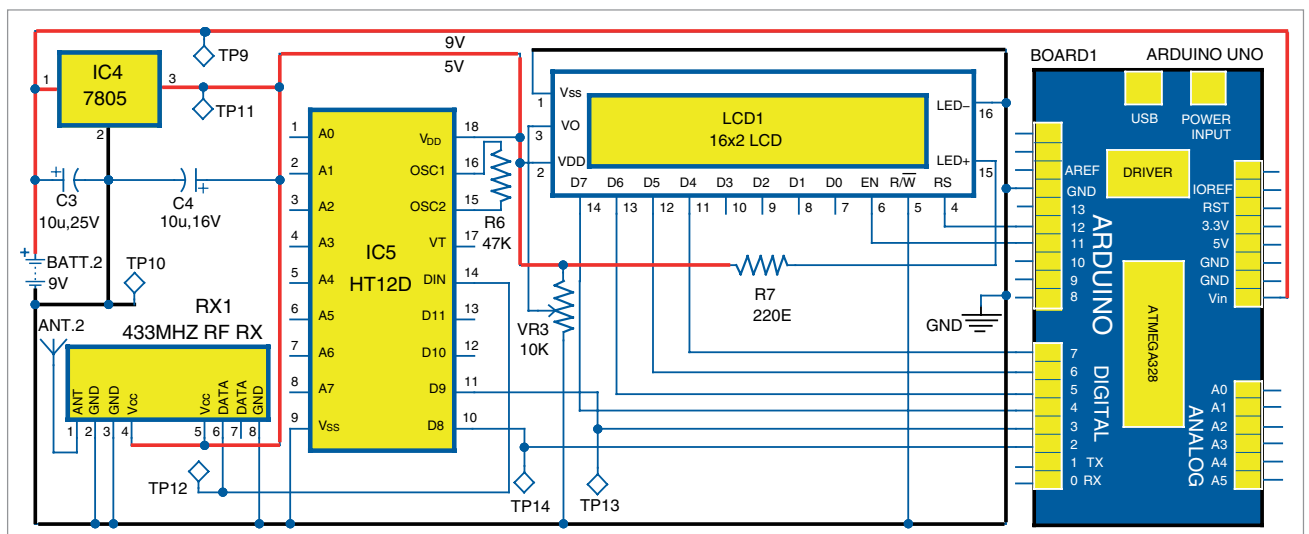


Fig. 6: Circuit diagram of receiver